OTHER YIELD MEASURES, CONVENTIONS, AND ACCOUNTING FOR EMBEDDED OPTIONS

Learning Outcome

- compare, calculate, and interpret yield and yield spread measures for fixed-rate bonds

Other Yield Measures and Conventions

The current yield, a simple measure equal to a bond's annual coupon divided by the flat price, was introduced in an earlier lesson. For example, if the BRWA 3.2% semiannual fixed-coupon five-year bond introduced earlier were trading at a price of 98.7 per $100 in face value at time *t*, its current yield would be

$$CY_t = \text{Annual coupon}_t/\text{Bond price}_t \qquad 2$$

$$= 3.2\%/0.987 = 3.242\%.$$

The current yield is a crude measure of return because it focuses solely on interest income, ignoring the frequency of coupon payments, interest on interest (time value of money), and accrued interest. In addition to collecting and reinvesting coupon payments, the investor has a gain if the bond is purchased at a discount and is redeemed at par value. The investor has a loss if the bond is purchased at a premium and is redeemed at par value.

In practice, the actual timing of cash flows is important for quoting and calculating bond yields-to-maturity. Consider from earlier the 30-year 4.625% annual coupon Romania eurobond maturing on 3 April 2049. Suppose that for settlement on 3 April 2026, the bond is priced at 97.3684 per 100 of par value to yield 4.75% quoted on an annual, actual/actual basis. Its coupon payment is scheduled for 3 April of each year. The yield calculation implicitly assumes that the payments are made on those exact dates. This calculation ignores the fact that several of those dates are not business days. For example, both 3 April 2027 and 3 April 2032 are Saturdays, so investors will receive coupon payments on the following Mondays, 5 April 2027 and 2032, respectively.

Yield measures that assume payments are made on scheduled dates and that do not account for weekends and holidays are quoted on what is called **street convention**. This assumption simplifies bond price and yield calculations and commonly is used in practice. A **true yield** that uses the actual payment dates (e.g., 5 April 2027 and 2032 in the Romania bond example), accounting for weekends and holidays, can also be computed. The true yield is never higher than the street convention yield because weekends and holidays delay the time to payment. The difference is typically small, no more than a basis point or two. Therefore, the true yield is not commonly used in practice.

Recall that corporate bond yields typically use the 30/360 day count convention. Sometimes, these yields are restated using an actual/actual day count, known as a **government equivalent yield**. The government equivalent yield on a corporate bond can be used to obtain the spread (difference) over a government bond yield.

EXAMPLE 4

Government Equivalent Yield for BRWA Five-Year Bond

In an earlier example, BRWA issued a 3.2% semiannual fixed-coupon five-year bond at par. As is standard for corporate bonds, the BRWA bond yield is calculated on a 30/360 day count basis. The comparable five-year US Treasury bond yields 2.3%,

Calculate the government equivalent yield for the BRWA bond and show the yield difference between the BRWA and government bonds based on the same day count basis.

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Solution:

A simple approximation for restating a corporate bond yield from the 30/360 day count basis to the actual/actual day count basis is to multiply the BRWA bond yield by 365/360.

$$Yield_{ACT/ACT} = \frac{365}{360} \times Yield_{30/360}.$$

$$Yield_{ACT/ACT} = \frac{365}{360} \times 0.032.$$

$$Yield_{ACT/ACT} = 3.244\%.$$

The bond equivalent yield for the BRWA bond is 3.2444%; hence, the yield-to-maturity difference between two bonds is 94.4 bps (= 3.2444% – 2.3%).

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Yield and Yield Spread Measures for Fixed-Rate Bonds

Sometimes the **simple yield** on a bond is quoted. It is the sum of the coupon payments plus the straight-line amortized share of the gain or loss, divided by the flat price. Simple yields are used mostly to quote Japanese government bonds, known as JGBs.

Exhibit 3:

Summary of Yield Conventions

Convention	Meaning
Actual/Actual (Act/Act)	Actual number of days from prior coupon payment to settlement date/number of days in a coupon period, assuming the actual number of days in a year. Typically used with government bonds.
30/360	Number of days from prior coupon payment to settlement date, assuming 30 days in a month/number of days in a coupon period, assuming 360 days in a year. Typically used with corporate bonds.
Street Convention	Yield measure that does not account for weekends and bank holidays and thus assumes cash flows are paid on their scheduled dates.
True Yield	Yield measure that accounts for weekends and bank holidays and thus assumes cash flows are paid after their scheduled dates. True yield is never higher than street convention yield due to the delay in time to payment.
Government Equivalent Yield	Yield measure that restates a yield-to-maturity based on a 30/360 day count to one based on an actual/actual day count. It is used to restate the YTM on a corporate bond to obtain the spread over the government YTM.
Simple Yield	Yield measure that is the sum of coupon payments plus the straight-line amortized share of the gain or loss, divided by the flat price. It is used mostly to quote Japanese government bonds (JGBs).

EXAMPLE 5

Comparing Yields for Different Periodicities

An analyst observes the following reported statistics for two bonds.

Coupon Payment Frequency	Quarterly	Semiannually
Years to Maturity	5 Years	5 Years
Price (per 100 of par value)	94	98.70
Current Yield	3.40%	2.53%
Yield-to-Maturity	4.548%	2.780%

1. Confirm the calculation of the current yield and yield-to-maturity for the two bonds.

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Solution:

Recall that the current yield is the annual coupon expressed as a percentage of a bond's flat price.

Current yield = $\frac{\text{Annual coupon}}{\text{Flat price}}$.

Current yield for Antelas AG bond:

$$CY = \frac{3.20}{94}.$$
$$CY = 0.0340 = 3.40\%.$$

Current yield for BRWA bond:

$$CY = \frac{2.50}{98.70}.$$
$$CY = 0.0253 = 2.53\%.$$

Recall that the yield-to-maturity is an internal rate of return, the discount rate r that equates the price of a bond to the present value of future cash flows.

$$PV = PMT_1/(1 + r)^1 + PMT_2/(1 + r)^2 + \ldots + (PMT_N + FV_N)/(1 + r)^N.$$

The yield-to-maturity can also be calculated using the YIELD and IRR functions in Microsoft Excel or Google Sheets.

Antelas AG Bond		BRWA Bond	
Settlement date	1/1/2000	Settlement date	1/1/2000
Maturity date	1/1/2005	Maturity date	1/1/2005
Coupon	0.032	Coupon	0.025
Price	94	Price	98.7
Redemption	100	Redemption	100
Coupon frequency	4	Coupon frequency	2
YIELD	**4.548%**	**YIELD**	**2.780%**

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Yield and Yield Spread Measures for Fixed-Rate Bonds

2. The analyst believes that the BRWA bond has less risk than the Antelas AG bond. How much additional compensation, in terms of basis points of yield-to-maturity, does a buyer of the Antelas AG bond receive for bearing this risk compared with the BRWA bond?

Hide Solution

Solution:

The yields-to-maturity on the two bonds have different periodicity assumptions. Therefore, the difference in the yields is *not* 177 bps (0.04548 – 0.02780). It is essential to compare the yields for the same periodicity to make a statement about relative value.

The Antelas AG bond yield-to-maturity of 4.548% for a periodicity of 4 converts to 4.574% for a periodicity of 2:

$APR_2 = 0.04574 = 4.574\%.$

When the yields-to-maturity for the Antelas AG and BRWA bonds are stated on a common periodicity, the additional yield that the buyer of the Antelas AG bond receives to compensate for its higher risk, compared to the BRWA bond, is (0.04574 – 0.02780) =179 bps.

Bonds with Embedded Options

If a fixed-rate bond contains an embedded option, other yield measures are used. Recall from an earlier lesson that these bond contingency provisions cannot be removed and traded separately from the bond. For example, a callable bond gives the issuer the right to buy the bond back from the investor. This occurs at specific prices on pre-determined dates after a call protection period during which the issuer is not allowed to exercise the call option.

For example, an earlier lesson on callable bonds introduced Vivivyu Inc. (VIVU) seven-year callable notes (see Exhibit 4).

Exhibit 4:

6.5% VIVU Seven-Year Callable Notes Term Sheet

6.5% Vivivyu Inc. (VIVU) Seven-Year Callable Notes (The "Notes")	
Prospectus Summary	
Issuer:	Vivivyu Incorporated
Settlement Date:	[T + 3 Business Days]
Maturity Date:	[Seven Years from Settlement Date]
Principal Amount:	US$ 400 million
Interest:	6.5% fixed coupon
Interest Payment:	Starting six months from [Settlement] to be paid semiannually with final payment on [Maturity]
Seniority:	The Notes are secured and unsubordinated obligations of VIVU and rank *pari passu* with all other secured and unsubordinated debt
Call Provision:	The Issuer may redeem some or all of the Notes on any Business Day before [Maturity] starting three years after [Settlement] based upon the Call Price Schedule as a percentage of the Principal Amount plus accrued interest
Call Price Schedule:	103.25% [Three to Four Years after Settlement]
	102.50% [Four to Five Years after Settlement]
	101.75% [Five to Six Years after Settlement]
	101.00% [Six to Seven Years after Settlement]

The VIVU bond may not be called for the first three years, but the following year it may be called at a price of $103.25 per $100 in face value. VIVU has the right to call the bond for the final four years based on a fixed price schedule. Investors facing call risk starting in Year 3 often estimate the lowest possible yield earned by calculating the sequence of yields corresponding to the bond's call dates.

Traditional yield-to-maturity measures assume that all cash flows occur as promised and therefore must be modified for callable bonds. Analysts must use alternative return measures that take the bond's call feature into account, such as the yield to the first call

$$(1+r)^1 \quad (1+r)^2 \qquad (1+r)^N$$

where

PV = the price of the bond

PMT = coupon payment per period

Call price = price at which a bond can be called on a given date

r = yield per period or market discount rate

N = number of evenly spaced periods to the date when a bond can be called at the call price

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Yield and Yield Spread Measures for Fixed-Rate Bonds

EXAMPLE 6

Yields-to-Call for the VIVU Bond

Assuming the call price schedule for the VIVU bond, the calculation of yields-to-first, -second, -third, -fourth, and -worst call and yield-to-maturity are as follows:

Yield-to-first call:

$$106.50 = \frac{3.25}{(1+r)^1} + \frac{3.25}{(1+r)^2} + \frac{3.25}{(1+r)^3} + \frac{3.25}{(1+r)^4} + \frac{3.25}{(1+r)^5} + \frac{103.25+3.25}{(1+r)^6}.$$

$$r = 0.025748 \times 2 = 0.0515.$$

Alternatively, the YIELD function in Microsoft Excel or Google Sheets can be used.

Yield-to-second call:

$$106.50 = \frac{3.25}{(1+r)^1} + \frac{3.25}{(1+r)^2} + \frac{3.25}{(1+r)^3} + \frac{3.25}{(1+r)^4} + \frac{3.25}{(1+r)^5} + \frac{3.25}{(1+r)^6} + \frac{3.25}{(1+r)^7} \frac{102.50+3.25}{(1+r)^8}.$$

$$r = 0.026237 \times 2 = 0.0525.$$

Yield-to-third call:

$$106.50 = \frac{3.25}{(1+r)^1} + \frac{3.25}{(1+r)^2} + \frac{3.25}{(1+r)^3} + \frac{3.25}{(1+r)^4} + \frac{3.25}{(1+r)^5} + \frac{3.25}{(1+r)^6} + \frac{3.25}{(1+r)^7} + \frac{3.25}{(1+r)^8} + \frac{3.25}{(1+r)^9} + \frac{101.75+3.25}{(1+r)^{10}}.$$

$$r = 0.02653 \times 2 = 0.0531.$$

Yield-to-fourth call:

$$106.50 = \frac{3.25}{(1+r)^1} + \frac{3.25}{(1+r)^2} + \frac{3.25}{(1+r)^3} + \frac{3.25}{(1+r)^4} + \frac{3.25}{(1+r)^5} + \frac{3.25}{(1+r)^6} + \frac{3.25}{(1+r)^7} + \frac{3.25}{(1+r)^8} + \frac{3.25}{(1+r)^9} + \frac{3.25}{(1+r)^{10}} + \frac{3.25}{(1+r)^{11}} + \frac{101.00+3.25}{(1+r)^{12}}.$$

$$r = 0.026811 \times 2 = 0.0536.$$

And finally, the yield-to-maturity assuming the bond is not called:

$$106.50 = \frac{3.25}{(1+r)^1} + \frac{3.25}{(1+r)^2} + \frac{3.25}{(1+r)^3} + \frac{3.25}{(1+r)^4} + \frac{3.25}{(1+r)^5} + \frac{3.25}{(1+r)^6} + \frac{3.25}{(1+r)^7} + \frac{3.25}{(1+r)^8} + \frac{3.25}{(1+r)^9} + \frac{3.25}{(1+r)^{10}} + \frac{3.25}{(1+r)^{11}} + \frac{3.25}{(1+r)^{12}} + \frac{3.25}{(1+r)^{13}} + \frac{100.00+3.25}{(1+r)^{14}}.$$

$$r = 0.026868 \times 2 = 0.0537.$$

In summary, the yield measures for the VIVU bond are as follows:

Yield-to-fourth call	5.362%
Yield-to-maturity	5.374%

The lowest of the sequence of yields-to-call and the yield-to-maturity is known as the **yield-to-worst**. For the VIVU bond, it is the yield-to-first call of 5.15%. The intent of this measure is to provide to the investor the most conservative measure for the rate of return.

The yield-to-worst is a commonly cited yield measure for fixed-rate callable bonds used by bond dealers and investors. However, a more precise approach is to use an option pricing model and an assumption about future interest rate volatility to value the embedded call option. The value of the embedded call option is added to the flat price of the bond to get the **option-adjusted price**. The investor bears the call risk (the bond issuer has the option to call), so the embedded call option reduces the value of the bond from the investor's perspective. The investor pays a lower price for the callable bond than if it were option-free. If the bond were non-callable, its price would be higher. The option-adjusted price is used to calculate the **option-adjusted yield**. The option-adjusted yield is the required market discount rate whereby the price is adjusted for the value of the embedded option. The value of the call option is the price of the option-free bond minus the price of the callable bond.

Question Set

1. Identify which of the following yield measures corresponds to the characteristics listed below.

Yield Measure	Characteristic
A. Simple Yield	1. Yield measure assuming one compounding period per year
B. Effective Annual Rate	2. Yield measure that restates a yield based on a 30/360 day count to one based on an actual/actual day count
C. Street Convention Yield	3. Yield measure that is the sum of coupon payments plus the straight-line amortized share of the gain or loss, divided by the flat price
D. Government Equivalent Yield	4. Assumes cash flows are paid on their scheduled dates, without accounting for weekends and bank holidays

Hide Solution

Solution:

Yield Measure	Characteristic
B. Effective Annual Rate	1. Yield measure assuming one compounding period in the year
D. Government Equivalent Yield	2. Yield measure that restates a yield-to-maturity based on a 30/360 day count to one based on an actual/actual day count
A. Simple Yield	3. Yield measure that is the sum of coupon payments plus the straight-line amortized share of the gain or loss, divided by the flat price
C. Street Convention Yield	4. Assumes cash flows are paid on their scheduled dates, without accounting for weekends and bank holidays

Yield Measure	Value
Yield-to-first call	0.000%
Yield-to-second call	0.157%
Yield-to-third call	0.939%
Yield-to-maturity	–0.385%

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Solution:

–0.3852%. The yield-to-worst is the lowest of the sequence of yields-to-call and the yield-to-maturity. The intent of yield-to-worst is to provide to the investor the most conservative assumption for the rate of return.

Use the following information compiled by a fixed-income analyst for two bonds, on the day both were issued, to answer Questions 3 and 4.

	Corporate Bond	Sovereign Bond
Annual coupon rate	4.00%	1.00%
Coupon frequency	Semiannual	Quarterly
Years to maturity	3 Years	3 Years
Price (per 100 of par)	91	102
Day-count basis	30/360	Actual/actual
Call price schedule:	101.25 1–2 Years after settlement	-

3. The current yield of the sovereign bond is approximately:

A. 140 bps *lower* than the current yield of the corporate bond.
B. 340 bps *lower* than the current yield of the corporate bond.
C. 140 bps *higher* than the current yield of the corporate bond.

Hide Solution

Solution:

B is correct. The current yield is the annual coupon expressed as a percentage of a bond's price. The current yield of the corporate bond is 440 bps, while the current yield of the sovereign bond is 98 bps, 342 bps lower.

4. The yield-to-first call of the corporate bond is closest to:

A. 2.15%.
B. 7.80%.
C. 15.00%.

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Solution:

C is correct. The yield-to-first call is the internal rate of return *r* for a callable bond assuming it is called at its earliest scheduled call date for its call price.

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \ldots + \frac{PMT + \text{Callprice}}{(1+r)^N}.$$

$$91 = \frac{2}{(1+r)^1} + \frac{101.25+2}{(1+r)^2}.$$

$$r = 0.0762 \times 2 = 0.1525.$$

The yield-to-first call for the bond is 15.25%.

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Yield and Yield Spread Measures for Fixed-Rate Bonds

What do you want to discuss... (required)

My YTM for Antela AG Bond in example 5 is 4.648 rather then 4.548. Anyone else?

Created 2 months ago by AHMAD SAID NJI TETGNU MBOUOMBOUO — 0 replies | Last Activity: 2 months ago

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DS

In question 6, how are they assuming the PV 106.5?

Created 5 months ago by Dustin Scott McElroy — 3 replies | Last Activity: 2 months ago

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JJ

Question 4 solution in the question set uses a call price of 101.50, but the information provided states 101.25.

Created 7 months ago by Jacobus Johannes Petrus Du Plessis — 3 replies | Last Activity: 3 months ago

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GA

Why in formula 4 VIVU exemple PV is not equal to 100 (par value)?

Created 3 months ago by Gabriel Aparecido Perez Jose — 0 replies | Last Activity: 3 months ago

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JB

can somone explain why in q4 answer is not b

Created 4 months ago by JIGAR BHARATBHAI RAJGOR — 4 replies | Last Activity: 4 months ago

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AH

In Question 4: 0,0762*2 equals 0,1524 and not 0,1512.

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